Logics & Symbolic AI – cheat sheet

1 Propositional logic

- **Syntax** = manipulating symbols; **semantics** = meaning.
- Formula: symbol p, constants $T, \bot$; $(\neg F)$; $(A \bullet B)$.
- Syntactic/semantic: ¬/Not, ∧/And, ∨/Or, ⊃/⇒, ⊂/⇐, ≡/⇔, ↑/Nand, ↓/Nor.
- **Valuation** $v : F \to \{True, False\}$, consistent **(4)**: **1)** $v(T) = True$ **2)** $v(\bot) = False$ **3)** $v(\neg F) = \text{Not } v(F)$ **4)** $v((A \bullet B)) = v(A) \blacksquare v(B)$.
- **Tautology**: $v(X) = True$ for every v.
- **Satisfiable** S: some v_0 makes all $X \in S$ True (SAT).
- X tautology $\iff \neg X$ not satisfiable.
- **Consequence** $S \models X$: every v satisfying S makes X True.
- **Key**: $S \models X \iff S \cup \{\neg X\}$ not satisfiable.
- Ex falso: $A, \neg A \in S \Rightarrow S \models X$ for all X.
- Monotony: $S \models X \Rightarrow S \cup \{A\} \models X$.
- Deduction: $S \cup \{X\} \models Y \iff S \models (X \supset Y)$.
- X taut. $\iff T \supset X$ taut., since $v(T \supset X) = v(X)$.

2 CNF & resolution

- **Negation normal form**: ¬ only before symbols.
- $[X_1..X_n]$ = generalized disjunction ($v([]) = False$).
- $\langle X_1..X_n \rangle$ = generalized conjunction ($v(\langle\rangle) = True$).
- **CNF** $\langle C_1,..,C_n \rangle$, each **clause** $C_i = [\ldots]$.

	rewrite	negation
∧	$X \wedge Y$	$\neg X \vee \neg Y$
∨	$X \vee Y$	$\neg X \wedge \neg Y$
⊃	$\neg X \vee Y$	$X \wedge \neg Y$
⊂	$X \vee \neg Y$	$\neg X \wedge Y$
↑	$\neg X \vee \neg Y$	$X \wedge Y$
↓	$\neg X \wedge \neg Y$	$X \vee Y$
⊅	$X \wedge \neg Y$	$\neg X \vee Y$
⊄	$\neg X \wedge Y$	$X \vee \neg Y$

- Disjunction: $[..P..] \to [..A,B..]$.
- Conjunction: $[..P..] \to [..A..], [..B..]$.
- $\neg\neg P \to P$, $\neg T \to \bot$, $\neg\bot \to T$.
- **Resolution**: $[A,X,B] + [C,\neg X,D] \to [A,B,C,D]$.
- Proof of X: start $[\neg X]$, end with $[]$.
- For $S \vdash X$: start $[S_1],..,[S_n],[\neg X]$.
- X tautology $\iff X$ has a resolution proof.
- Prolog `A :- B` $= [A, \neg B]$.

Ex. $((A \supset B) \supset (A \supset C))$
$\langle [\neg(A \supset B), A \supset C] \rangle \to \langle [A \wedge \neg B, A \supset C] \rangle$
$\to \langle [A, \neg A, C], [\neg B, \neg A, C] \rangle \to \langle [\neg B, \neg A, C] \rangle$

- **Backward** chaining: from goal (resolution).
- **Forward** chaining: from assumptions (Hilbert, modus ponens $\frac{X \quad X \supset Y}{Y}$).
- $\models$ semantic (truth tables); $\vdash$ syntactic (proofs).
- **Sound**: $S \vdash F \Rightarrow S \models F$.
- **Complete**: $S \models F \Rightarrow S \vdash F$.

3 First-order logic

- **Terms**: variables, constants, functors $f(t_1..t_n)$.
- **Atomic formula**: predicate $p(t_1..t_n)$, arity n.
- Formulas: atomic, $\neg F$, $(F \circ G)$, $(\forall x)F$, $(\exists x)F$.
- **Functor or predicate?** Inside an argument ⇒ functor. At formula level (¬, ∧, ⊃, quantifier) ⇒ predicate.
- **Free** v **(4 rules)**: **1)** occurs in atomic F **2)** in $(\diamond v')F$ iff free in F and $v \neq v'$ **3)** in $\neg F$ iff in F **4)** in $F_1 \circ F_2$ iff in F_1 or F_2.
- **Closed** formula: no free variable.

Semantics

- Domain $D \neq \emptyset$; interpretation I **(3)**: **1)** $c^I \in D$ **2)** $f^I : D^n \to D$ **3)** p^I n-ary relation.
- Assignment A: $v^A \in D$. Model $M(D,I)$.
- **Attitude**: $p(t_1..)^{I,A} = True$ iff $(t_1^{I,A},..) \in p^I$.
- $(\forall x)F$: True for all B equal to A save for x; $\exists$: for one.
- **Valid**: true in every model (all A).
- F valid $\iff \{\neg F\}$ not satisfiable.
- Valid: $((\forall x)p(a,x)) \supset p(a,a)$.
- Not valid: $(\forall x)(p(a,x) \supset p(a,a))$, $(\forall x)(\exists y)p(x,y)$.

Prenex (rename variables first)

- $\neg(\exists v)A \equiv (\forall v)\neg A$; $\neg(\forall v)A \equiv (\exists v)\neg A$.
- $((Qv)A \wedge B) \equiv (Qv)(A \wedge B)$, either side, $Q \in \{\forall, \exists\}$.
- $((\forall v)A \supset B) \equiv (\exists v)(A \supset B)$ (flips!).
- $((\exists v)A \supset B) \equiv (\forall v)(A \supset B)$ (flips!).
- $(A \supset (Qv)B) \equiv (Qv)(A \supset B)$ (kept).

Skolemization

- $..(\exists x_k)..F \to F(x_k / f(x_1..x_{k-1}))$, new functor.
- Arity = number of ∀ before it (0 = constant).
- Preserves satisfiability; put ∃ first if you can.
- $(\forall x)(m(x) \supset (\exists y)s(x,y)) \to (\forall x)(m(x) \supset s(x, f(x)))$.

Prove F valid by resolution (6 steps)

1. $\neg F$ in prenex form (rename variables).
2. Skolemize; **3)** drop ∀; **4)** CNF (optional).
5. Resolution + **unification** $(P(x), \neg P(c)$: $x = c)$.
6. $[] \Rightarrow \{\neg F\}$ unsat $\Rightarrow F$ valid.

- Prop.: sound, complete, **decidable**.
- FOL: sound, complete, **semi-decidable** (valid ⇒ proof found in finite time; not valid ⇒ may never stop).

4 Natural deduction

$$\text{Ax}\ \frac{}{\Gamma, p \vdash p} \qquad \neg\text{I}\ \frac{\Gamma, p \vdash \bot}{\Gamma \vdash \neg p}$$

$$\neg\text{E}\ \frac{\Gamma \vdash p \quad \Gamma \vdash \neg p}{\Gamma \vdash \bot} \qquad \text{Abs}\ \frac{\Gamma, \neg p \vdash \bot}{\Gamma \vdash p}$$

$$\bot\text{E}\ \frac{\Gamma \vdash \bot}{\Gamma \vdash p} \qquad \to\text{I}\ \frac{\Gamma, p \vdash q}{\Gamma \vdash p \to q}$$

$$\to\text{E}\ \frac{\Gamma \vdash p \to q \quad \Gamma \vdash p}{\Gamma \vdash q} \qquad \wedge\text{I}\ \frac{\Gamma \vdash p \quad \Gamma \vdash q}{\Gamma \vdash p \wedge q}$$

$$\wedge\text{Eg}\ \frac{\Gamma \vdash p \wedge q}{\Gamma \vdash p} \qquad \wedge\text{Ed}\ \frac{\Gamma \vdash p \wedge q}{\Gamma \vdash q}$$

$$\vee\text{Ig}\ \frac{\Gamma \vdash p}{\Gamma \vdash p \vee q} \qquad \vee\text{Id}\ \frac{\Gamma \vdash q}{\Gamma \vdash p \vee q}$$

$$\vee\text{E}\ \frac{\Gamma \vdash p \vee q \quad \Gamma, p \vdash r \quad \Gamma, q \vdash r}{\Gamma \vdash r}$$

$$\forall\text{I}\ \frac{\Gamma \vdash F}{\Gamma \vdash \forall x F} \qquad \forall\text{E}\ \frac{\Gamma \vdash \forall x F(x)}{\Gamma \vdash F[t/x]}$$

$$\exists\text{I}\ \frac{\Gamma \vdash F(t)}{\Gamma \vdash \exists x F(x)} \qquad \exists\text{E}\ \frac{\Gamma \vdash \exists x F \quad \Gamma \vdash F \to G}{\Gamma \vdash G}$$

- ∀I: x not free in Γ. ∃E: x not free in Γ, G.

Ex. $\vdash (A \to B) \to ((B \to C) \to (A \to C))$
$\Gamma = \{A \to B, B \to C, A\}$: Ax gives A, $A \to B$;
→E: B; →E with $B \to C$: C; then →I ×3.
Curry–Howard term: $\lambda f.\lambda g.\lambda x.g(fx)$.

5 Planning: STRIPS

- **State** describes the world; **actions** change it.
- Planning = action sequence from initial to goal state.
- Predicates/Variables Capitalized; constants lowercase.
- `Move(X,Y)` Pre: `At(X), Level(low)`
  Post: `!At(X), At(Y)`
- STRIPS solver = backtracking: semi-decidable, may explore pointless strategies.
- **Sussman anomaly**: `On(a,b)`, `On(b,c)` interfere in either order.

6 Planning as SAT (Kautz & Selman 1992)

1. Add **time** as the last argument: `Move(a,b,1)`.
2. **Ground** every predicate (all argument values).
3. Grounded clause → boolean: `move_a_b_12`, `U-1-2`.
4. Write the **(5 kinds)** of constraints below.

- **1) Initial** ($t = 0$): every grounded fluent, + or ¬ (1 model).
- **2) Goal** ($t = N$): only goal facts (many models).
- **3) Problem**: e.g. table always clear, one action per step.
- **4) Action**: $a_t \Rightarrow (p_t \wedge c_{t+1})$.
- **5) Explanatory frame axioms (2 per fluent)**:
  - $(\neg f_t \wedge f_{t+1}) \Rightarrow \bigvee$ actions at t that add f.
  - $(f_t \wedge \neg f_{t+1}) \Rightarrow \bigvee$ actions at t that delete f.
- SAT is decidable but N is fixed: increase N until satisfiable.
- Lab: `Object(X,T), Source(Y,T), Destination(Z,T).`
- MDP: $p(s'|s,a)$, reward r, policy $\pi : s \mapsto a$, maximize $\sum_{t=0}^{\infty} \gamma^t E[r_t]$.

7 Formal Concept Analysis

- **Context** $K = (X, Y, I)$, $I \subseteq X \times Y$ (objects × attributes).
- $A^\uparrow = \{y \in Y \mid \forall x \in A : (x,y) \in I\}$ (shared attributes).
- $B^\downarrow = \{x \in X \mid \forall y \in B : (x,y) \in I\}$ (objects having B).
- $\emptyset^\uparrow = Y$, $\emptyset^\downarrow = X$.
- **Concept** (A,B) **(2, both!)**: **1)** $A^\uparrow = B$ **2)** $B^\downarrow = A$. A extent, B intent.
- **Galois connection (4)**: **1)** $A_1 \subseteq A_2 \Rightarrow A_2^\uparrow \subseteq A_1^\uparrow$ **2)** $B_1 \subseteq B_2 \Rightarrow B_2^\downarrow \subseteq B_1^\downarrow$ **3)** $A \subseteq A^{\uparrow\downarrow}$ **4)** $B \subseteq B^{\downarrow\uparrow}$.
- **Closure operator (3)**: **1)** extensive $A \subseteq T(A)$ **2)** monotone **3)** idempotent $T(T(A)) = T(A)$.
- $(\cdot)^{\uparrow\downarrow}$ closes extents; $(\cdot)^{\downarrow\uparrow}$ closes intents.
- Always a concept: $(B^\downarrow, B^{\downarrow\uparrow})$ and $(A^{\uparrow\downarrow}, A^\uparrow)$.
- **Order**: $(A_1,B_1) \preceq (A_2,B_2) \iff A_1 \subseteq A_2 \iff B_2 \subseteq B_1$.
- **Meet** $\left(\bigcap X_t, (\bigcup Y_t)^{\downarrow\uparrow}\right)$; **Join** $\left((\bigcup X_t)^{\uparrow\downarrow}, \bigcap Y_t\right)$.

Intersection method (all concepts) (5 steps)
1. List each object intent $\{x\}^\uparrow$.
2. Add Y (bottom intent).
3. Add pairwise intersections until nothing new.
4. Extent of each intent B: $B^\downarrow$.
5. Order by extent inclusion; draw only immediate edges.

Dual: intersect attribute extents $\{y\}^\downarrow$, close with $^\uparrow$.

- **Labels**: attribute y at $\{y\}^\downarrow$ (highest); object x at $\{x\}^\uparrow$ (lowest).
- Intent = labels above; extent = labels below.
- **NextClosure**: intents in lectic order $y_1 < .. < y_n$.
  - Try $y_i \notin B$ from last to first.
  - $C = ((B \cap \{y_1..y_{i-1}\}) \cup \{y_i\})^{\downarrow\uparrow}$.
  - Accept if C adds no attribute before y_i.
  - Start $\emptyset^{\downarrow\uparrow}$; stop at Y. $O(|X||Y|^2|C|)$.
- **Clarified**: no identical rows/columns (isomorphic lattice).
- **Reduced**: no object/attribute derivable from others.
- **Implication** $Z_1 \Rightarrow Z_2$ valid $\iff Z_1^\downarrow \subseteq Z_2^\downarrow$.
- Counterexample: object in $Z_1^\downarrow \setminus Z_2^\downarrow$.
- **Armstrong (3)**: **1)** reflexivity **2)** augmentation **3)** transitivity.
- **Basis (2)**: **1)** complete **2)** non-redundant.
  Duquenne–Guigues: pseudo-intents $P \Rightarrow P^{\downarrow\uparrow}$, minimal.

Slide ex.: table of concepts
Context: x_1:$y_1y_2y_3y_4$, x_2:$y_1y_3y_4$, x_3,x_4:$y_2y_3y_4$, x_5:y_1.

	extent A	intent B	below
C_1	$x_1x_2x_3x_4x_5$	$\emptyset$	(top)
C_2	$x_1x_2x_3x_4$	y_3y_4	C_1
C_3	$x_1x_3x_4$	$y_2y_3y_4$	C_2
C_4	$x_1x_2x_5$	y_1	C_1
C_5	x_1x_2	$y_1y_3y_4$	C_2, C_4
C_6	x_1	$y_1y_2y_3y_4$	C_3, C_5

NextClosure order: $C_1, C_2, C_3, C_4, C_5, C_6$. Clarify: $x_3 = x_4$, $y_3 = y_4$.

8 Association rules

- $\sigma(X)$ = number of transactions containing X.
- $s(X) = \sigma(X)/|T|$; frequent if $s(X) \geq minSup$.
- $s(X \Rightarrow Y) = \sigma(X \cup Y)/|T|$.
- $c(X \Rightarrow Y) = \sigma(X \cup Y)/\sigma(X)$.
- **Lift** $= c/s(Y) = \frac{s(X \cup Y)}{s(X)s(Y)}$: $= 1$ indep., > 1 positive.
- Co-occurrence, NOT causality.
- **Apriori**: support anti-monotone; infrequent ⇒ supersets infrequent.
- Pass k: join L_{k-1}, prune, scan, keep $\geq minSup$.
- Rules from L: $A \Rightarrow L \setminus A$, $c = s(L)/s(A)$.

FCA vs AR
- FCA implication: deterministic, holds for all objects.
- AR: probabilistic (support, confidence, lift).
- Implication = AR with confidence 100%.
- FCA: small exact data. AR: large noisy data.

9 Fuzzy logic

- $\neg x = 1 - x$; Yager $\neg_w(x) = (1 - x^w)^{1/w}$.
- **T-norm (4)**: **1)** commutative **2)** monotone **3)** associative **4)** neutral $T(a,1) = a$. So $T(a,0) = 0$.
- **(4 common)**: min (largest); product ab; Łuk. $\max(0, a+b-1)$; drastic (smallest).
- Weak contradiction: $T(a, 1-a) \leq 0.5$.
- **T-conorm** $\bot(a,b) = 1 - T(1-a, 1-b)$ **(4)**: comm., monotone, assoc., identity $\bot(a,0) = a$.
- max (smallest); $1-(1-a)(1-b)$; $\min(a+b, 1)$; drastic (largest).
- **S-impl.** $(\neg x \vee y)$: $\max(1-x, y)$, $1-x(1-y)$, $\min(1-x+y, 1)$.
- **R-impl.** $R_T(x,y) = \sup\{z : T(z,x) \leq y\}$:
  - $= 1$ if $x \leq y$;
  - else min: y; product: y/x; Łuk.: $1-x+y$.
- $I(\neg a) = 1-a$, $I(a \wedge b) = T$, $I(a \vee b) = \bot$, $I(a \Rightarrow b) = R_T$.
- **Simple positive inference**: $P_1 \wedge .. \wedge P_n \Rightarrow_\phi C$.
  - Satisfied if $C \geq \phi(P_1..P_n)$.
  - Solution **(3)**: **1)** ≥ initial values **2)** all rules satisfied **3)** no output can be smaller.
  - Iterate rules in any order until all satisfied.
  - With negation: may have no solution / oscillate.
- **Fuzzy set** $\mu : U \to [0,1]$ **(3 sets)**: support $\mu > 0$; core $\mu = 1$; α-cut $\mu \geq \alpha$.
- α-capping $\min(\alpha, \mu(x))$.
- **Equality relation (3)**: **1)** commutative **2)** reflexive $E(x,x) = 1$ **3)** transitive $T(E(x,y), E(y,z)) \leq E(x,z)$.
- Fuzzification $\mu(x) = \sup\{E(x,y) : y \in A\}$.
- **Defuzzification (3 methods)**:
  - First of Maxima $\min\{c \mid \mu(c) \max\}$: safe, ignores other maxima.
  - Mean of Maxima: may land on a non-maximum.
  - Median: equal mass on both sides; may also miss.
- **Mamdani (4 steps)**: **1)** fuzzify inputs **2)** $\alpha = T(\ldots)$ **3)** cap

C at α 4) defuzzify.

10 MAX SAT & PSL

- **Weighted MAX SAT**: maximize total weight of satisfied formulas.
- Generalizes SAT (NP-complete): no poly algorithm in general.
- Relaxation (Łuk.): clause degree $\min(\sum_i x_i, 1)$.
- Negative literal contributes $1 - y(s)$.
- **PSL**: max $\sum_c w(c) \min(\sum_{s+} y(s) + \sum_{s-} (1 - y(s)), 1)$.
- Constraints $y(s) \le v$. **Convex** $\Rightarrow$ polynomial (hit and run).

11 Theorem proving & λ-calculus

- **Hilbert's program** (3): 1) consistent 2) complete 3) decidable.
- **Gödel 1**: consistent system with arithmetic is incomplete.
- **Gödel 2**: it cannot prove its own consistency.
- λ-terms: variable; abstraction $\lambda x.y$; application fx (left assoc.).
- β: $(\lambda x.M)N \to M[N/x]$; rename bound variables to avoid capture.
- **Strongly normalizing**: every reduction terminates.
- $(\lambda x.xx)(\lambda x.xx)$ never terminates; cannot be typed.

0	$\lambda f.\lambda x.x$	$n+1$	$\lambda n.\lambda f.\lambda x.f(nfx)$
1	$\lambda f.\lambda x.fx$	$m+n$	$\lambda m.\lambda n.\lambda f.\lambda x.mf(nfx)$
n	$\lambda f.\lambda x.f^n x$	mn	$\lambda m.\lambda n.\lambda f.\lambda x.m(nf)x$
True	$\lambda a.\lambda b.a$	$\neg$	$\lambda p.p$ False True
False	$\lambda a.\lambda b.b$	$\wedge;\vee$	$\lambda p.\lambda q.pqp$; $\lambda p.\lambda q.ppq$

- **Typed**: $\lambda x.y : \sigma \to \tau$; $fx : \tau$ needs $f : \sigma \to \tau$, $x : \sigma$.
- Simply typed λ is strongly normalizing (Curry).
- **Curry–Howard**: formula = type, proof = program.
  - $\to$I = abstraction; $\to$E (modus ponens) = application.
  - $\wedge$ = product, $\vee$ = sum, $\top$ = unit, $\bot$ = empty.
  - $\forall$ = dependent product, $\exists$ = dependent sum.
- Non-constructive proofs use excluded middle $P \vee \neg P$.

λ sheet

- (1) y; (2) yz; (4) y.
- (3) $\lambda w.\lambda x.y$ (rename bound y to w).
- (5) $\lambda y.yy(yy)$.
- $\lambda n.n(\lambda x.\text{false})$true: is n zero?
- $\lambda m.\lambda n.nm$: exponent m^n.
- Predecessor: $\lambda u.x$ swallows the first $f \Rightarrow f^{n-1}x$.

12 Description logics

- Variable-free, decidable fragments of FOL.
- **KB = TBox** (concepts, roles) **+ ABox** (named individuals).
- $\mathscr{AL}$ (6): $\top$, $\bot$, $\neg A$ (atomic only), $C \sqcap D$, $\forall R.C$, $\exists R.\top$.
- $\mathscr{U}$: $\sqcup$; $\mathscr{E}$: $\exists R.C$; $\mathscr{C}$: $\neg C$; $\mathscr{N}$: $\le nR$, $\ge nR$.
- $\mathscr{AL} \subseteq \mathscr{ALE} \subseteq \mathscr{ALC} \subseteq \mathscr{SHOIN} \subseteq \mathscr{SROIQ}$ (OWL 2 DL).

Semantics $\mathscr{I} = (\Delta^I, \cdot^I)$

- $A^I \subseteq \Delta^I$, $R^I \subseteq \Delta^I \times \Delta^I$.
- $(C \sqcap D)^I = C^I \cap D^I$, $(\neg C)^I = \Delta^I \setminus C^I$.
- $(\exists R.C)^I = \{x \mid \exists y, (x,y) \in R^I \wedge y \in C^I\}$.
- $(\forall R.C)^I = \{x \mid \forall y, (x,y) \in R^I \to y \in C^I\}$.

Translation τ_x to FOL

- $\tau_x(C) = C(x)$; $\sqcap \to \wedge$; $\sqcup \to \vee$; $\neg \to \neg$.
- $\tau_x(\exists r.C) = \exists y, r(x,y) \wedge \tau_y(C)$.
- $\tau_x(\forall r.C) = \forall y, r(x,y) \to \tau_y(C)$.
- $C \sqsubseteq D$: $\forall x(\tau_x(C) \to \tau_x(D))$.
- ABox: $C(a)$, $r(a,b)$.

Reasoning: TBox (4), ABox (4)

- TBox model: satisfies all $C \sqsubseteq D$, $C \equiv D$.
- Satisfiable: $C^I \neq \emptyset$ in some model of $\mathscr{T}$.
- Subsumption $\mathscr{T} \models C \sqsubseteq D$: $C^I \subseteq D^I$ in all models.
- Disjoint: $C^I \cap D^I = \emptyset$ in all models.
- $C \sqsubseteq D \iff C \sqcap \neg D$ unsatisfiable.
- Disjoint $\iff C \sqcap D$ unsat.; unsat $\iff C \sqsubseteq \bot$.
- ABox **consistent**: a model of $\mathscr{A}$ and $\mathscr{T}$ exists.
- **Instance**: $\mathscr{A} \models C(a) \iff \mathscr{A} \cup \{\neg C(a)\}$ inconsistent.
- **Retrieval**: all a with $C(a)$. **Realization**: most specific C.

Inconsistent ABox

- $\mathscr{T}$: Father $\sqsubseteq$ Man, Mother $\sqsubseteq$ Woman, Man $\sqcap$ Woman $\sqsubseteq \bot$.
- $\mathscr{A}$: Mother(mary), Father(mary).
- $mary^I \in Woman^I \cap Man^I = \emptyset$: inconsistent.

Ontologies & OWL

- Ontology (4 words): 1) formal 2) explicit specification of a 3) shared 4) conceptualization.
- OWL restrictions (6): some ($\exists$), only ($\forall$), hasValue, $= n$, $\le n$, $\ge n$.
- Properties (5): inverse, transitive, symmetric, functional, inverse functional.
- Functional: $P(X,Y) \wedge P(X,Z) \Rightarrow Y = Z$.
- Protégé: EquivalentTo = necessary and sufficient.
- Modus ponens $\frac{A, A \to B}{B}$; modus tollens $\frac{\neg B, A \to B}{\neg A}$.

13 DL exam-style exercises

Prove $\forall r.(A \sqcap B) \sqsubseteq \forall r.A \sqcap \forall r.B$

- Take $x \in (\forall r.(A \sqcap B))^I$ and y with $(x,y) \in r^I$.
- Then $y \in A^I \cap B^I \subseteq A^I$, so $x \in (\forall r.A)^I$.
- Same for B, so x is in the intersection.
- FOL: $\forall x[\forall y(r \to A \wedge B) \to \forall y(r \to A) \wedge \forall y(r \to B)]$ is valid.
- The converse holds too ($\equiv$).
- $\exists r.(A \sqcap B) \sqsubseteq \exists r.A \sqcap \exists r.B$, *not* the converse.

Is Pedro a SuccessfulMan?

- SM $\equiv \neg$F $\sqcap$ InBusiness $\sqcap \exists$married.Lawyer $\sqcap \exists$child.(Uni $\sqcup$ Works).
- ABox: Pedro:$\neg$F, Pedro:InBusiness, Mary:Lawyer, John:Works, married(Pedro,Mary), child(Pedro,John).
- Yes: Mary witnesses $\exists$married; John witnesses $\exists$child.
- $\equiv$ is sufficient; with only $\sqsubseteq$ it would not follow.
- Method: check each conjunct; witness for each $\exists$; all fillers for each $\forall$.

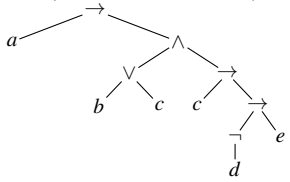
Interpretation: $\Delta^I = \{John, Mary\}$, $\text{Human}^I = \Delta^I$, $\text{hasChild}^I = \{(John, Mary)\} \Rightarrow (\exists \text{hasChild.Human})^I = \{John\}$.

14 Example exam – solutions

Propositional logic

- $F = a \to ((b \vee c) \wedge (c \to (\neg d \to e)))$.
- False iff $a = 1$ and ($b = c = 0$, or $c = 1, d = e = 0$).
- $abcde$: 10000, 10001, 10010, 10011, 10100, 11100.
- CNF: $\langle [\neg a, b, c], [\neg a, \neg c, d, e] \rangle$.

Decomposition tree (root = main connective):



First-order logic

- Predicates (formula level): $a/2$, $b/3$, $d/1$, $e/1$.
- Functions (inside terms): $c/2$, $f/2$, $g/1$.
- Free: x, y (in $a(x,y)$, $e(f(x,g(y)))$) and z (never bound).

Planning ($pX1_3$ = posX-1-3, $L2_3$ = L-2-3)

- Init: $pX1_0 \wedge \neg pX2_0 \wedge \neg pX3_0 \wedge pY1_0 \wedge \neg pY2_0 \wedge \neg pY3_0$.
- Goal: $pX2_5 \wedge pY3_5$.
- Frame axioms, posX at $t = 3$:
  - $(\neg pX1_3 \wedge pX1_4) \to L2_3$; $(pX1_3 \wedge \neg pX1_4) \to R1_3$
  - $(\neg pX2_3 \wedge pX2_4) \to R1_3 \vee L3_3$
  - $(pX2_3 \wedge \neg pX2_4) \to R2_3 \vee L2_3$
  - $(\neg pX3_3 \wedge pX3_4) \to R2_3$; $(pX3_3 \wedge \neg pX3_4) \to L3_3$
- Up, $t = 2$:
  - $U1_2 \to pY1_2 \wedge \neg pY1_3 \wedge pY2_3$
  - $U2_2 \to pY2_2 \wedge \neg pY2_3 \wedge pY3_3$
  - $\neg U3_2$ (precondition $Y < 3$ false)

Fuzzy

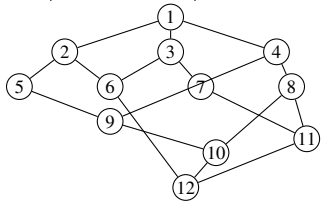
- Average: not a T-norm: $T(0,1) = 0.5 \neq 0$.
- $(a \to b) \vee c$: $a \le b \Rightarrow 1$; $\max(1, 0.1) = 1$.
- $a \wedge b \to c$, $c \to a$, $b = .3$, $c = .1$, $a = 0$: $a = .1$, then all satisfied.
- Weighted MAX SAT: NP-hard. With $[0,1]$ values: PSL, polynomial.
- PSL $\{a\}[1], \{\neg a, \neg b\}[2], \{b\}[3]$: $a = 0$, $b = 1$, value 5.

FCA (ME, W, M, I, R, Fe, E)

- Aragorn = Boromir: clarify. ME is a full column.
- Object intents: Fr{ME,R,Fe}, Ar=Bo{ME,W,Fe}, Ga{ME,W,M,I,Fe}, Le{ME,W,I,Fe}, Sa{ME,M,I,R,E}, Go{ME,R,E}.
- Their pairwise intersections add {ME}, {ME,Fe}, {ME,R}, {ME,I}, {ME,M,I}.

Table of concepts (every intent also contains ME)

#	intent	extent	below
1	ME	all 7	(top)
2	Fe	Fr Ar Ga Le Bo	1
3	R	Fr Sa Go	1
4	I	Ga Le Sa	1
5	W Fe	Ar Ga Le Bo	2
6	R Fe	Fr	2, 3
7	R E	Sa Go	3
8	M I	Ga Sa	4
9	W I Fe	Ga Le	4, 5
10	W M I Fe	Ga	8, 9
11	M I R E	Sa	7, 8
12	all 7 attributes	$\emptyset$	6, 10, 11



- Fe $\to$ c2; Magic $\to$ c8; Warrior $\to$ c5.
- $(\{Ga, Sa\}, \{M, I\})$: no, $\{Ga, Sa\}^\uparrow = \{ME, M, I\}$.
- $(\{Fr, Ar\}, \{Fe\})$: no, $\{Fe\}^\downarrow$ has 5 objects.
- $\{W, Fe\} \to \{I\}$: invalid (Aragorn, Boromir); $s = 2/7$, $c = 1/2$.
- $\{R, Fe\} \to \{\neg E\}$: holds (Frodo); $s = 1/7$, $c = 1$.

15 Proof templates

- **Galois**: $Y \subseteq A^\uparrow \iff A \subseteq Y^\downarrow$.
- **Antitone**: m shared by all of $X' \supseteq X$ is shared by X.
- $\alpha\beta\alpha = \alpha$: $\supseteq$ extensive; $\subseteq$ antitone on $X \subseteq X^{\uparrow\downarrow}$.
- **Meet is a concept**: $(Y_1 \cup Y_2)^\downarrow = Y_1^\downarrow \cap Y_2^\downarrow = X_1 \cap X_2$.
- **Anti-monotone**: $Y_1 \subseteq Y_2 \Rightarrow \sigma(Y_1) \ge \sigma(Y_2)$.
- **min largest**: $T(a,b) \le T(a,1) = a$, same for b.
- **English→FOL**: "all A are B" $\forall x(A \supset B)$; "some" $\exists x(A \wedge B)$.
- Friends of John know one of his children: $(\forall x)(fr(x,j) \supset (\exists y)(ch(y,j) \wedge kn(x,y)))$.

16 Worked slide examples

Prop. resolution $((A \supset B) \wedge (B \supset C)) \supset \neg(\neg C \wedge A)$

- Goal line: $[\neg(\ldots)]$ gives $[(A \supset B) \wedge (B \supset C)]$, $[\neg C \wedge A]$.
- Lines: $[\neg A, B]$, $[\neg B, C]$, $[\neg C]$, $[A]$.
- Resolve: $[B]$, then $[C]$, then $[]$. Valid.

FOL resolution $(\exists x)(A \wedge \neg P(x)) \supset \neg(\forall x)P(x)$

- Negate: $(\exists x)(A \wedge \neg P(x))$ and $(\forall x)P(x)$.
- Skolemize: A, $\neg P(c)$; drop $\forall$: $P(x)$.
- Unify $x = c$: $[]$. Valid.

Prenex $(\exists x)(\forall y)R(x,y) \supset (\forall y)(\exists x)R(x,y)$

- Rename: $\ldots \supset (\forall u)(\exists v)R(v,u)$.
- $(\forall x)(\exists y)(\forall u)(\exists v)(R(x,y) \supset R(v,u))$ (antecedent flips).
- Skolemize: $y = f(x)$, $v = g(x,u)$.

Fuzzy (slides): product $R(.9,.2) = .2/.9 \approx .22$; $I(.22 \wedge .9) = .198$. Mamdani: HOT .9, CROWDED .6 $\Rightarrow \alpha = .6$.

MAX SAT $\{s\}[7]$, $\{\neg s, \neg b\}[8]$, $\{b\}[10]$

- Worlds (s,b): $(0,1) = 18$, $(1,0) = 15$, $(1,1) = 17$, $(0,0) = 8$.

Forward chaining B, C true; prove H

- R1 BDE→F, R2 DG→A, R3 CF→A, R4 C→D, R5 D→E, R6 A→H, R7 B→X, R8 XC→A.
- First rule: D, E, F, A, H. Last rule: X, A, H.

17 Exam recipes (step by step)

- **When is F false?** Main connective first: $X \to Y$ false iff $X = 1, Y = 0$; split cases; list all rows.
- **CNF** (3): 1) remove $\supset$ 2) push $\neg$ inward 3) distribute $\vee$ over $\wedge$; drop clauses with $p, \neg p$.
- **Unsat KB, 2 ways**: truth table; resolution to $[]$.
- **Frame axioms**: for each grounded fluent, 2 axioms (becomes true, becomes false); RHS = actions with that effect.
- **Action constraint**: one per grounding; impossible precondition $\Rightarrow \neg a_t$.
- **Concept check** (2): $A^\uparrow = B$? $B^\downarrow = A$? Closure of m: $\{m\}^{\downarrow\uparrow}$.
- **Implication** $Z_1 \to Z_2$: does every object of $Z_1^\downarrow$ have Z_2? Then s, c.
- **T-norm?** Test the 4 axioms; first try $T(a,1) = a$.
- **R-implication**: $x \le y \Rightarrow 1$; else T-norm's formula.
- **Fuzzy inference**: raise each conclusion to ϕ(premises); repeat until stable.
- **PSL**: write each clause term; case split where $\min(\cdot, 1)$ changes; maximize.
- **DL instance**: unfold the definition; witness for each $\exists$; requires $\equiv$.
- **DL subsumption**: take $x \in C^I$, show $x \in D^I$; or $C \sqcap \neg D$ unsat.
- **λ reduction**: leftmost outer redex first; rename before substituting.